

The statistics of soaring flight

Displacement scaling across terrain, circuit types and wing classes

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OFFSITE
2026

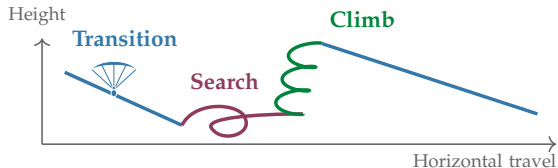
EConophysics Lab

CFM

ILB Institut
Louis
Bachelier

Soaring is transport with an energy budget

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Phase segmentation: HMM of Vilpellet et al.

Time-averaged MSD

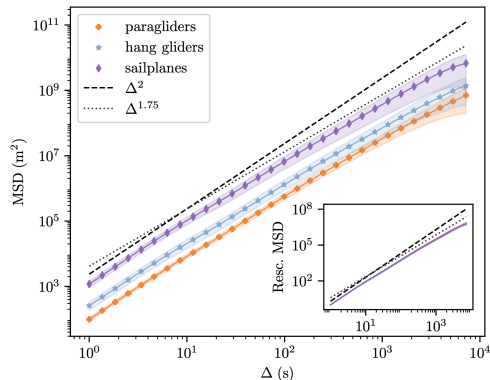
- 1 Along each flight f , of duration T_f

$$m_f(\tau) = \frac{1}{T_f - \tau} \int_0^{T_f - \tau} |\mathbf{x}_f(t + \tau) - \mathbf{x}_f(t)|^2 dt$$

- 2 Over the N flights: $M_2(\tau) = \frac{1}{N} \sum_f m_f(\tau)$

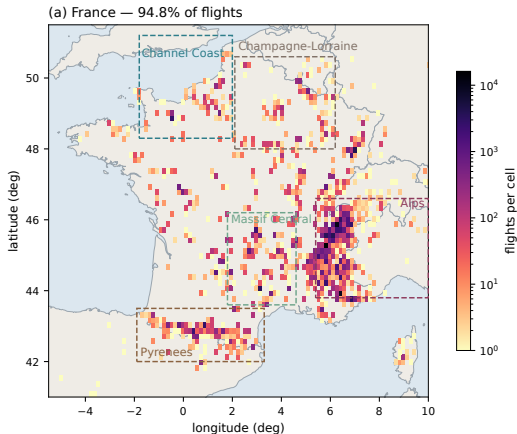
$M_2(\tau) \propto \tau^{2H}$ with $H \approx 0.88$: superdiffusive over three decades, 10^1 – 10^4 s

Vilpellet et al.: French flights, 2016–2021



A large archive of free flight

3



FFVL distance competition

186,052

Paraglider tracklogs
1999-2026

6,716

Hang-glider tracklogs
2001-2026

Regional boxes

- Channel Coast
- Champagne-Lorraine
- Massif Central
- Alps
- Pyrenees

IGC trajectories + metadata: time, position, altitude, site, circuit and wing class

France map: 148,081 paraglider and 5,894 hang-glider flights. Colour: flights per 0.15° cell.

From raw tracklogs to the fixed cohort C_{10000}

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Preprocessing

Clean the recorded flight



Selection and coordinates



Uniform time grid



Savitzky–Golay smoothing



Retained (of downloads)

156,406

paragliders, 84%

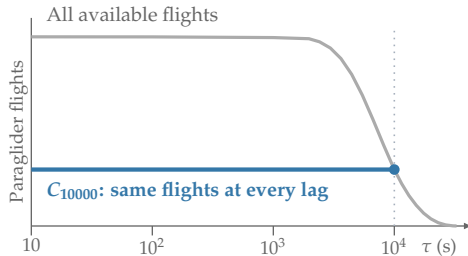
6,094

hang gliders, 91%



Fixed cohort C_{10000}

Segments of at least 12,500 s (≈ 3.5 h)



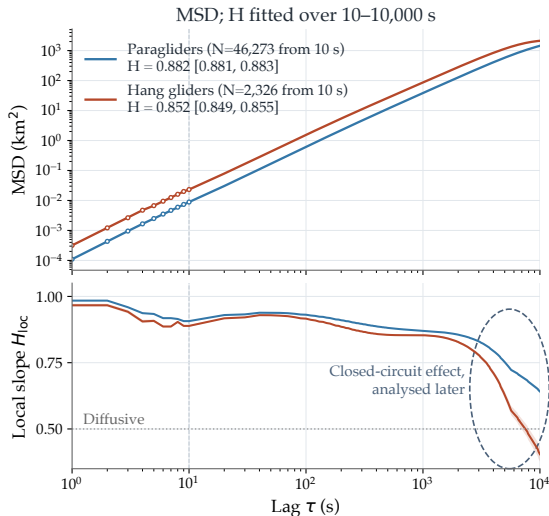
In C_{10000} (of retained)

46,273

paragliders, 30%

2,326

hang gliders, 38%



As Vilpellet et al. (slide 2): mean along each flight, then over flights

$$M_2(\tau) = \left\langle \overline{|\mathbf{x}_f(t+\tau) - \mathbf{x}_f(t)|^2}^t \right\rangle_f$$

$\overline{\cdot}^t$ mean over origins t along flight f

$\langle \cdot \rangle_f$ mean over the N flights, equal weights

Slightly different from Vilpellet et al.

- Vilpellet et al., 2016–2021: $H \approx 0.88$ for both
- Here: $H = 0.88$ paragliders, 0.85 hang gliders

Which flight characteristics shape MSD and H ?

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Circuit

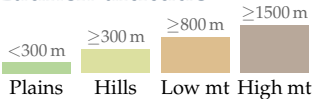


open



closed

Launch altitude



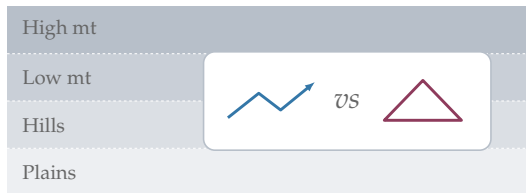
Wing class

Beginners EN A/B/C
Experts EN D/CCC

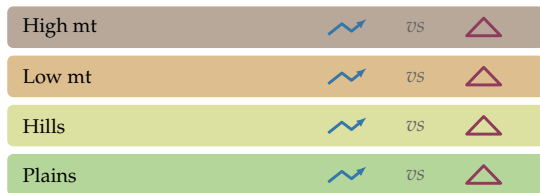
Weather

not observed

Marginal across all altitude classes



Stratified within each altitude class

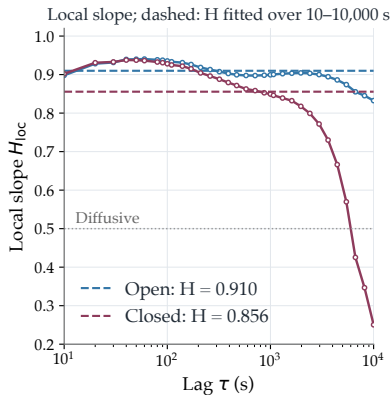
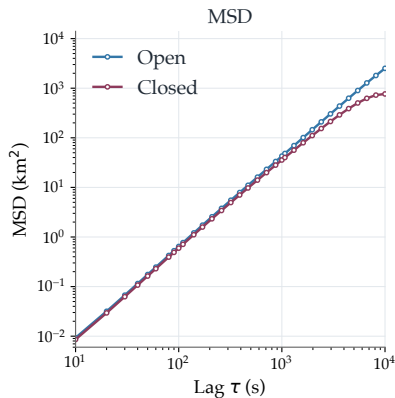


First compare groups overall, then within the same altitude class.

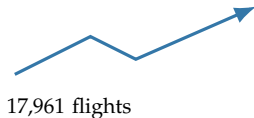
Altitude class set by z_0 , the GNSS altitude of the first fix after trimming.

Open vs closed: how do MSD and H differ?

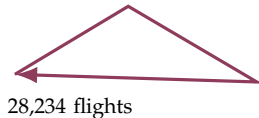
7



Open circuits



Closed circuits

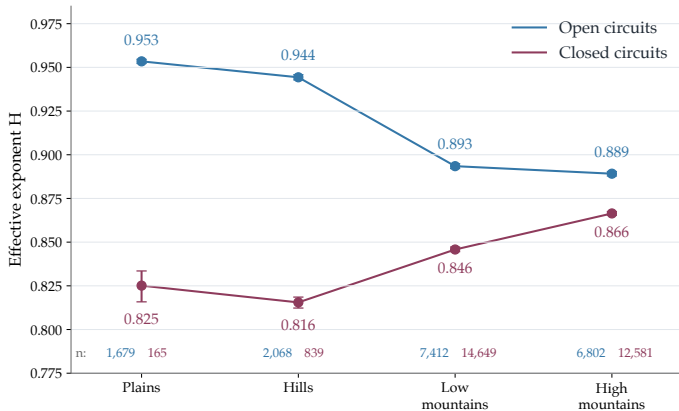


$$H_{\text{open}} - H_{\text{closed}} = 0.054 [0.053, 0.056] > 0 \quad (90\% \text{ paired interval})$$

Slower closed-route MSD growth is consistent with turns and returns.

Does the open–closed H gap change with altitude?

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Open: higher H in lowlands

- Lowlands: no deviations for lift
- Mountains: deviations to known lift

Closed: higher H in mountains

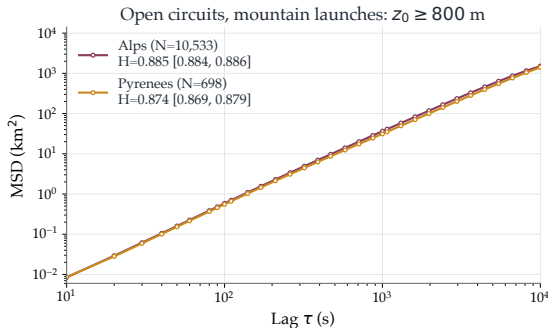
- Lowlands: turn back any time
- Mountains: way home follows ridges

Open > closed everywhere: gap 0.13 in lowlands, 0.02–0.05 in mountains.

Mountain terrain constrains both circuits, so the gap narrows.

Open circuits: how does H differ by region?

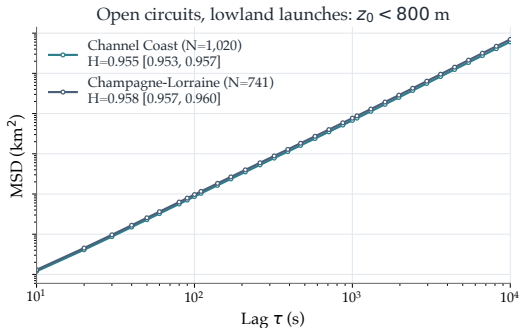
9



Mountains: Alps above Pyrenees

Alps – Pyrenees

0.011 [0.006, 0.016]



Lowlands: almost equal

Channel Coast – Champagne-Lorraine

–0.003 [–0.006, –0.001]

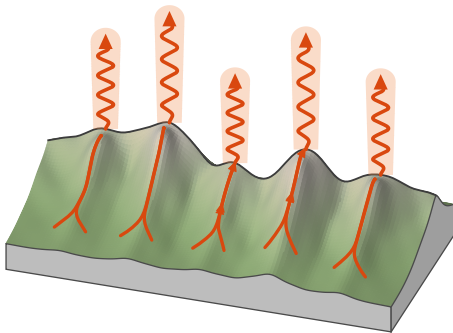
H separates the two mountain ranges more clearly than the two plains.

Why does H separate the ranges but not the plains?

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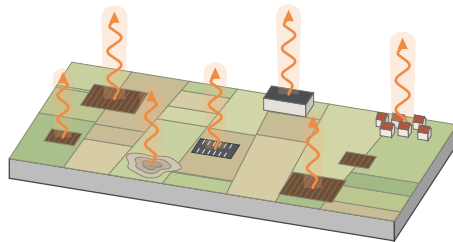
Mountains: lift follows the slopes

Strong thermals climb sunny spurs to the crest.



Lowlands: scattered triggers

Fields, quarry, asphalt, roofs: random spots, weaker lift.



Mountain lift traces each range's relief; lowland lift is scattered at random.

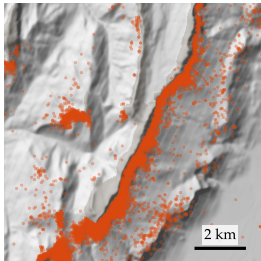
So the Alps and Pyrenees can differ in H , while two plains with similar trigger statistics cannot.

Where do climbs cross one horizontal plane?

11

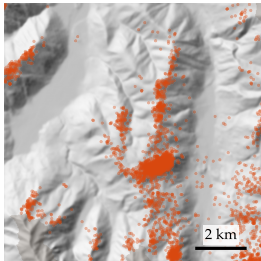
Mountains: narrow bands on the relief

Alps



$H = 1,741$ m, 7,890 flights
half the crossings in **0.7%**

Pyrenees



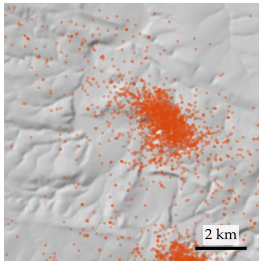
$H = 1,868$ m, 2,352 flights
half the crossings in **0.2%**

● climb crossing of the plane ■ terrain above the plane

Bold: share of the cell holding half the crossings; uniform points would need 11%.

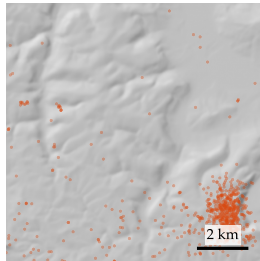
Lowlands: broad clouds

Channel Coast



$H = 747$ m, 1,454 flights
half the crossings in **1.7%**

Champagne-Lorraine



$H = 830$ m, 500 flights
half the crossings in **1.1%**

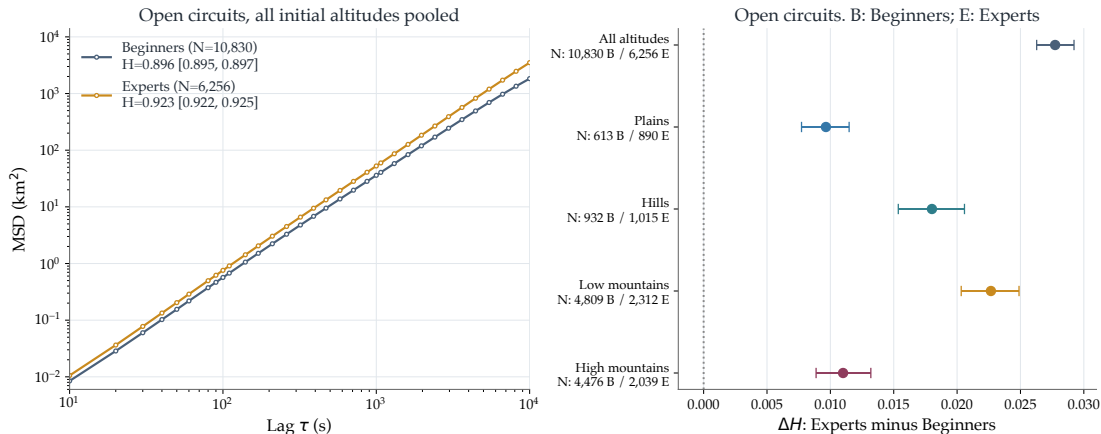
Plane: $H = \text{mean terrain of the } 10 \times 10 \text{ km cell} + 600 \text{ m}$

At one height: narrow bands in the mountains, broad clouds in the lowlands.

No cell is close to uniform: crossings also mark where pilots launch.

Open circuits: expert–beginner H gap by altitude

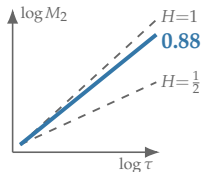
12



Higher H for experts in every class; the gap depends on altitude.

Wing class is a proxy. Nominal 90% cluster-bootstrap CIs (Appendix).

Scaling



Strongly superdiffusive

- Paragliders $H = 0.88$
- Hang gliders $H = 0.85$

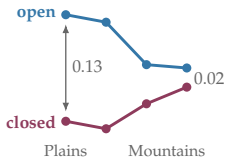
Route and wing



Open routes spread faster

Experts spread faster

Terrain



Mountains narrow the gap

- Open-route H falls
- Closed-route H rises

Lift geography



Mountain lift hugs ridges

- H2: on ridges at 600 m
- P1: spread out at 600 m

Strongly superdiffusive everywhere; route and terrain set how strongly.

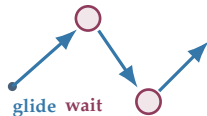
Segmentation



Revise Vilpellet's HMM

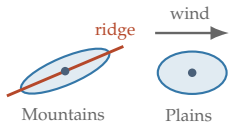


Stochastic model



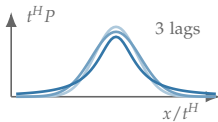
Which model fits best?

Anisotropy



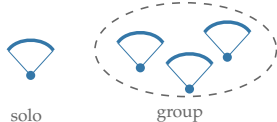
Ridge axis or wind

Distribution scaling



Do distributions collapse?

Solo vs group flight



Does group flight change H ?

Thank you
for your attention!

Appendix: clean the recorded flight

↔ slide 4 16

1 GNSS altitude

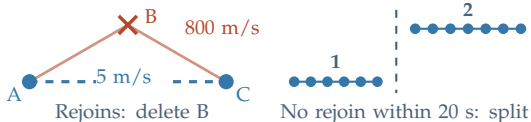
GNSS height → Trajectory $z(t)$

Barometer → Cleaning checks

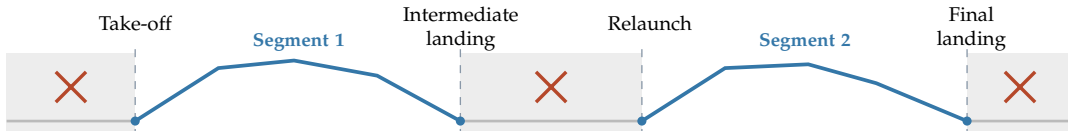
GNSS: absolute height without weather drift.
Same high-frequency noise as the barometer.

2 Fix-level cleaning

One fix = (t, φ, λ, z) : time, latitude, longitude, height
Delete an unreachable fix only if the track rejoins without it; otherwise split the flight.



3 Airborne interval



Trim before take-off and after landing.
Stops are not labeled: we infer them from position and altitude.

4 Flight selection

Minimum flight thresholds

Time ≥ 40 min

1.13% excluded ($N = 191,108$)

Path ≥ 20 km

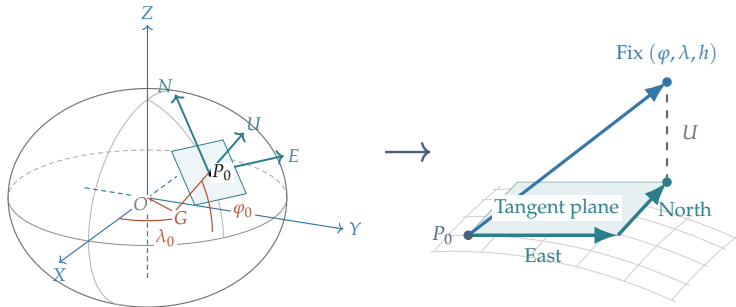
0.09% excluded ($N = 188,942$)

Altitude range ≥ 600 m

2.68% excluded ($N = 188,768$)

5 Coordinates

Latitude, longitude, height $\rightarrow$ Earth-centred (X, Y, Z)
 $\rightarrow$ local east-north distances in metres



Earth model: WGS84 ellipsoid (GPS reference)

P_0 : first trimmed fix; fixed for the flight. $z = h_{\text{GNSS}}$, not U .

6a Gaps: bridge or split

Gap g between consecutive fixes:

- $g \leq g_{\max}$: bridge it.
- $g > g_{\max}$: split into segments.

$$g_{\max} = \min(10\Delta t, \max(20 \text{ s}, 2\Delta t))$$

Example: $\Delta t = 5 \text{ s}$, $g_{\max} = 20 \text{ s}$

Short gap
15 s: bridge



Long gap
30 s: split



● Raw fix
(jittered)

● Grid point

○ Flagged as
reconstructed

6b Time grid: jitter removal

Per segment: grid $t_k = t_0 + k\Delta t$, native Δt .

- Interpolate E , N , z at every t_k .
- No fix within $\Delta t/2$ of t_k : flag it as reconstructed.



Keep a segment only if it lasts $\geq 90 \text{ s}$ and $\leq 10\%$ of its grid is reconstructed.

No segment kept: flight dropped. 20,798 paragliders, 193 hang gliders: the largest loss.

7 Savitzky-Golay fit

Fit a cubic to E , N , z in a moving window, within each segment.

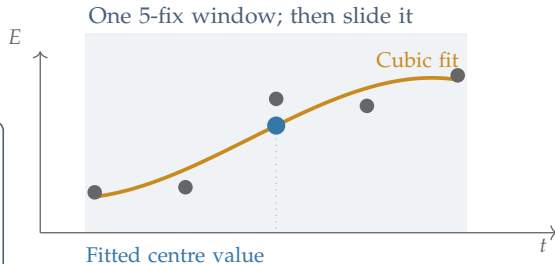
Window: odd, ≥ 5 fixes.

Velocity and acceleration: derivatives of the fitted cubic

$$v_i = P'_i(0), \quad a_i = P''_i(0), \quad u = t - t_i$$

Noise std, for an independent error σ per fix:

	velocity	acceleration
Fitted cubic	$0.95 \sigma / \Delta t$	$0.53 \sigma / \Delta t^2$
Fix-to-fix differences	$1.41 \sigma / \Delta t$	$2.45 \sigma / \Delta t^2$



Retained flights after all seven stages (share of downloaded logs)

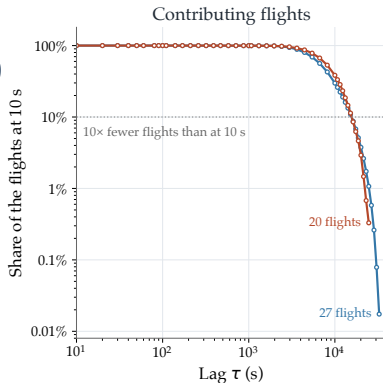
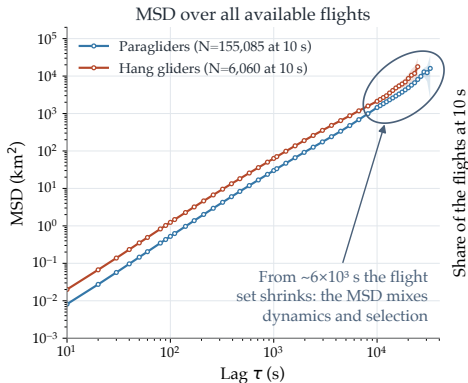
156,406 **paragliders** 84.1%

6,094 **hang gliders** 90.7%

Schematic; the cubic fit is exact for the synthetic fixes shown.

Appendix: which flights contribute at each lag?

All available flights: a flight enters at lag τ if one of its segments has $\tau \leq 0.8 T_s$ ($\geq 1/5$ of it left as origins for m_f).



Desiderata

- 1 The same flights at every lag
- 2 The longest possible lag range
- 3 $N \approx$ of the same order as at 10 s

At long lags only long flights remain, and their number falls steeply.

Any lag-dependent statistic then mixes dynamics with a changing set of flights.

Appendix: can we afford a fixed population?

↔ slide 4 21

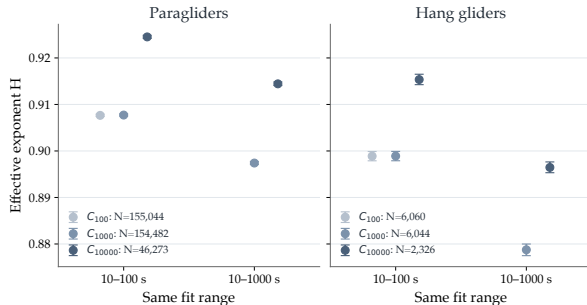
C_{10000} : select once for the largest lag

$\tau_{\max} = 10,000 \text{ s} \leq 0.8 T_s$: $T_s \geq 12,500 \text{ s}$
Only these segments enter, at every lag.

Advantages of C_{10000}

- 1 Same flights and segments, 10 to 10,000 s, with no loss of flights beyond $6 \times 10^3 \text{ s}$
- 2 Three decades of lag; at $\tau_{\max}$ each segment still offers at least 251 origins for m_f
- 3 Sample still large, same order as at 10 s:
46,273 paragliders (30%)
2,326 hang gliders (38%)

Cost of selecting longer flights



Nested cohorts, identical fit ranges, paired bootstrap draws.

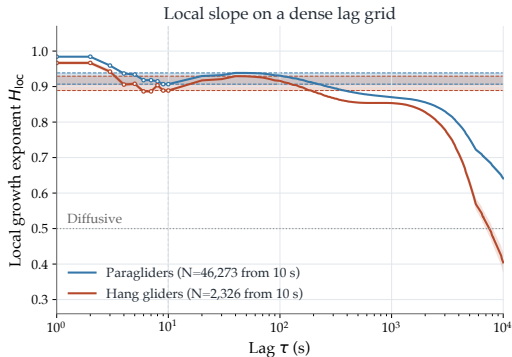
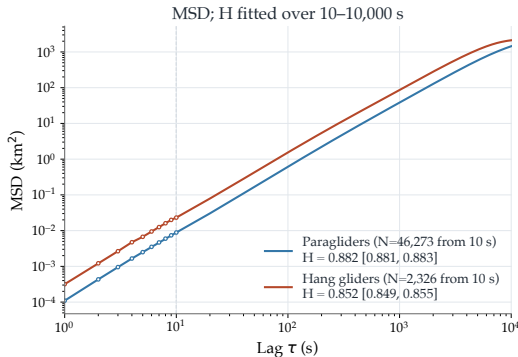
Long flights raise H by 0.016–0.018 on matched lag ranges; origins still decrease with lag.
About 2% in H buys the same flights at every lag over three decades.

Displacement growth from 1 s to 10,000 s

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C_{10000} : changing support below 10 s; the same flights at every lag from 10 to 10,000 s.

$M_2(\tau) \propto \tau^{2H}$: $H = \frac{1}{2}$ diffusion $H = 1$ ballistic motion



$$H_{\text{PG}} = 0.882 [0.881, 0.883] > H_{\text{HG}} = 0.852 [0.849, 0.855] \text{ over } 10\text{--}10,000 \text{ s}$$

Nominal 90% bootstrap CIs; clusters sampled with replacement. Clustering remains too fine; bands likely too narrow.

1 Group flights

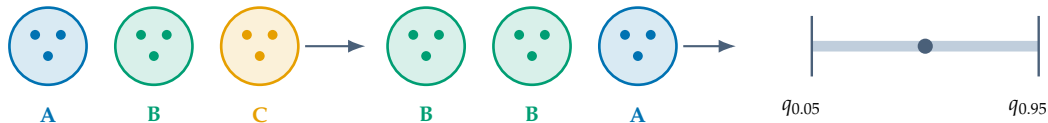
Take-off site $\times$ calendar day.
Aim: independent groups.

2 Resample groups

Sample with replacement.
Keep each group together.

3 Recompute

1,000 replicates.
Take the 5th–95th percentiles.



Current clusters are too small to absorb all shared flight conditions.

Intervals likely underestimate uncertainty; clustering is still being refined

Flight archive

FFVL Coupe Fédérale de Distance: IGC tracks, dates, declared sites, circuits and wing classes.

parapente.ffvl.fr/cfd/liste delta.ffvl.fr/cfd/liste

Maps and terrain

[OpenTopoMap](#): regional relief and contours (OSM / SRTM, CC-BY-SA).

[IGN / Géoplateforme](#): local Plan IGN backgrounds.

[IGN RGE ALTI](#): local terrain and derived crests.

Locator and analysis

[Natural Earth](#): France outline, public domain.

MSD, intervals and crossings: this thesis's prepared pipeline.

[Code, figure scripts and recorded source provenance](#)